

Ana Lorena Ortiz Loyola

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Data science student (Tecnológico de Monterrey, expected 2027) with a quantitative foundation in statistics, numerical optimization, and cryptographic complexity analysis, plus hands-on financial-services experience through a data engineering internship at Deutsche Bundesbank (AnaCredit) involving automated data-quality and audit-style verification of regulatory data. Works in a structured, independent style with strong interest in applying quantitative methods to financial markets.

Core Competencies

- **Quantitative & Financial Analysis:** Statistics, numerical optimization, predictive modeling, interest in financial markets
- **Financial Services & Audit-Adjacent Experience:** Data quality verification, migration testing, regulatory reporting (Deutsche Bundesbank AnaCredit)
- **Programming:** Python (Pandas, NumPy, Scikit-Learn), R, SQL, Java, C++
- **Machine Learning:** Scikit-Learn, Keras, TensorFlow, Random Forest, SVM, K-Means
- **Independent Research:** Cryptographic complexity and security-parameter analysis, contributing to a scientific publication
- **Cloud & Tools:** Azure Cognitive Services, AWS, Git

Professional Experience

- **Deutsche Bundesbank (AnaCredit)** **Frankfurt am Main, Germany**
Data Engineer / Intern *04/2026–07/2026*
 - Developed automated migration tests (row counts, referential integrity, aggregate consistency) using Pandas DataFrames to verify identical business logic after data migration – audit-style validation of regulatory reporting data
 - Built automated error detection identifying statistical deviations (duplicate IDs, value-range violations) in large-scale reported banking data, with automated Jira ticketing for detected issues

Projects

- **Tecnológico de Monterrey**
Lattice-Based Cryptography Research 10/2024–02/2025
 - Implemented and analyzed the complexity of lattice-based cryptographic algorithms (LWE, NTRU) in Python; contributed to a scientific publication analyzing security parameters and runtime behavior
- **Tecnológico de Monterrey**
Predictive Modeling – Healthcare Optimization 10/2024–12/2024
 - Analyzed 25+ years of national health data; built and validated predictive models (Random Forest, Neural Networks) and K-Means clustering
- **Personal Project (GitHub), built with Claude Code**
Banking Data Pipeline – Data Cleaning & Audit Log 05/2026
 - Built an end-to-end pipeline for simulated bank transactions (150 accounts): normalized SQLite database with foreign-key integrity checks and a fully auditable cleaning log
- **Tecnológico de Monterrey**
Air Quality Analysis Mexico City – SVM Classification 04/2023–06/2023
 - Built SVM models (linear and RBF kernels) to classify critical ozone levels from citywide NO_x/O₃ data

Education

- **Tecnológico de Monterrey** **Mexico**
BSc in Data Science 2022–2027
Grade: 1.3 (Academic Excellence Scholarship; DAAD KOSPIE Scholarship). Topics: statistics, numerical optimization, machine learning & AI, cryptography & security.
- **Universität Göttingen** **Germany**
Exchange Semester 2025–2026

Languages

- German (C1), English (C1, TOEFL iBT 105/120), Spanish (native).

Publications

- Contributed to a scientific publication on lattice-based cryptography (LWE, NTRU security parameter and runtime analysis), with Prof. A. F. De Abiega L'Eglise, Tecnológico de Monterrey (citation pending publication).

Honors and Awards

- **Academic Excellence Scholarship** – Tecnológico de Monterrey.
- **DAAD KOSPIE Scholarship**.
- **National Physics Olympiad** – Mexico (2020).
- **Honorable Mention, Physics Olympiad** (2021).

References

- Available upon request.